HACETTEPE UNIVERSITY
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING
ELE 432: Advanced Digital Design
Experiment 3 / Preliminary Work

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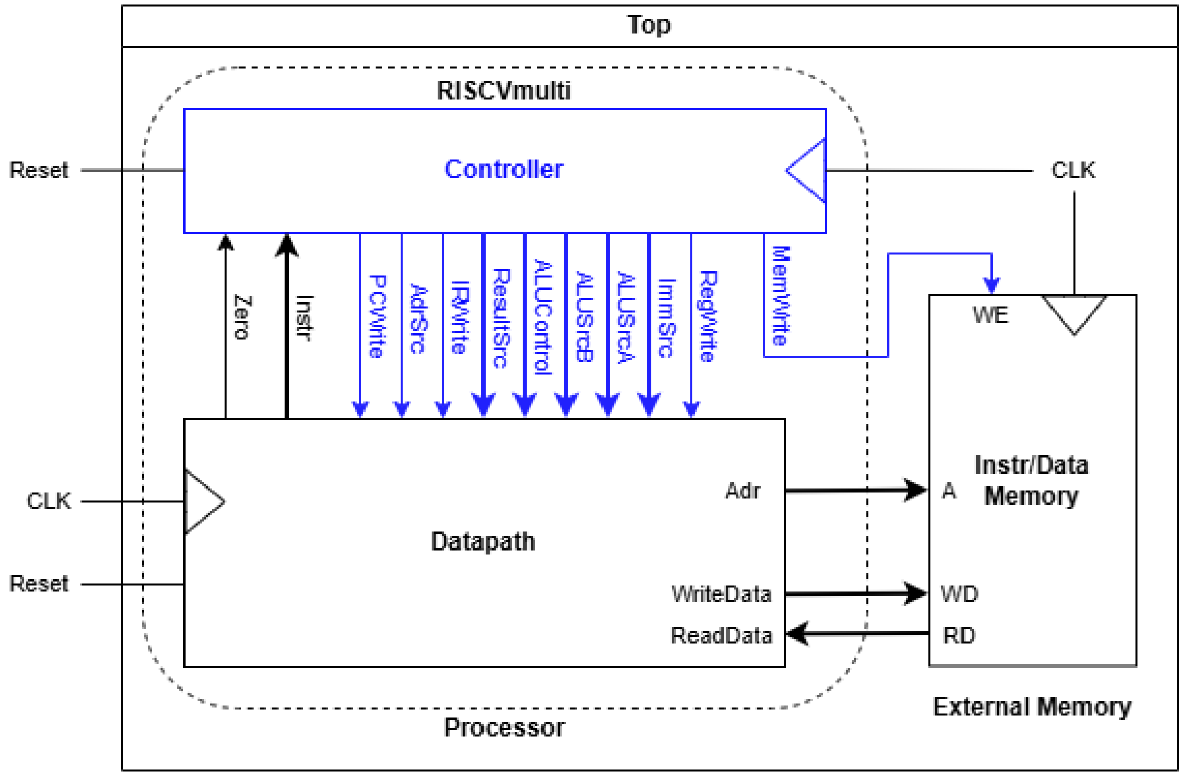
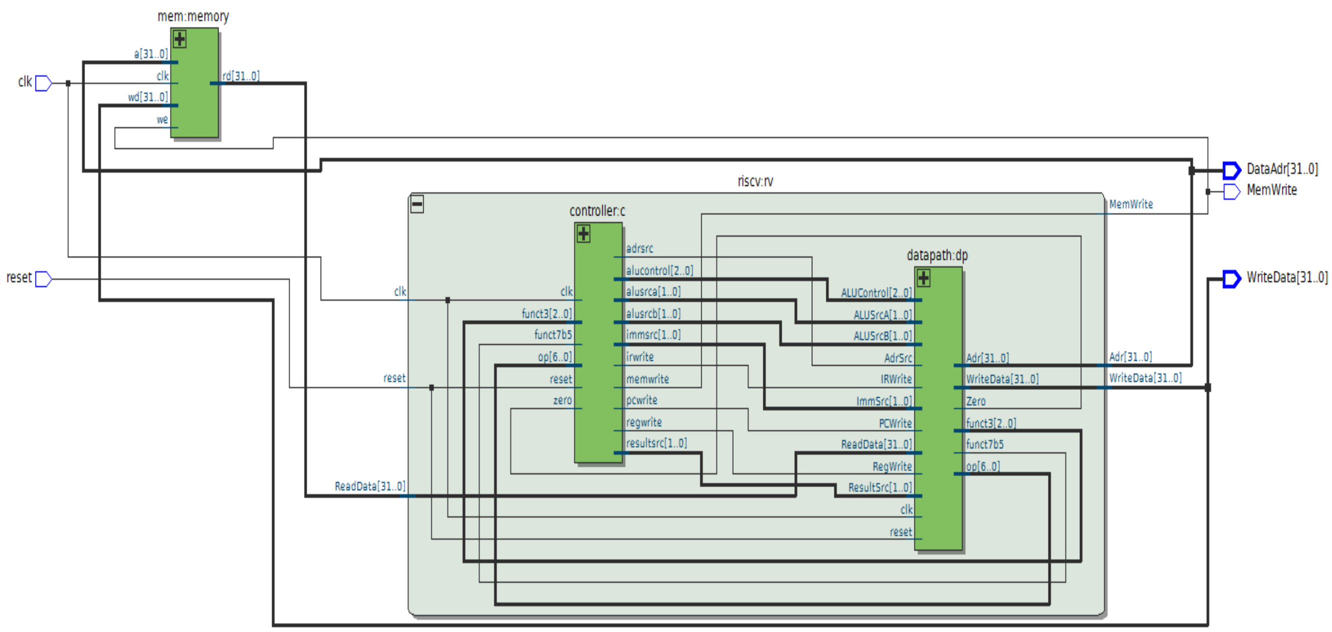
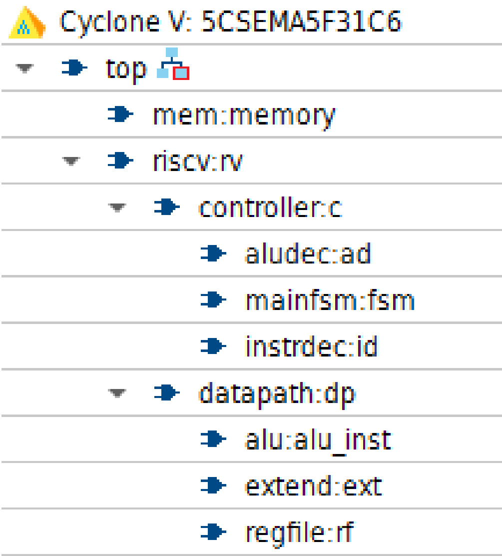
1) This lab took me around 6 hours. Since I have already finished the controller circuit for the previous homework.

2)





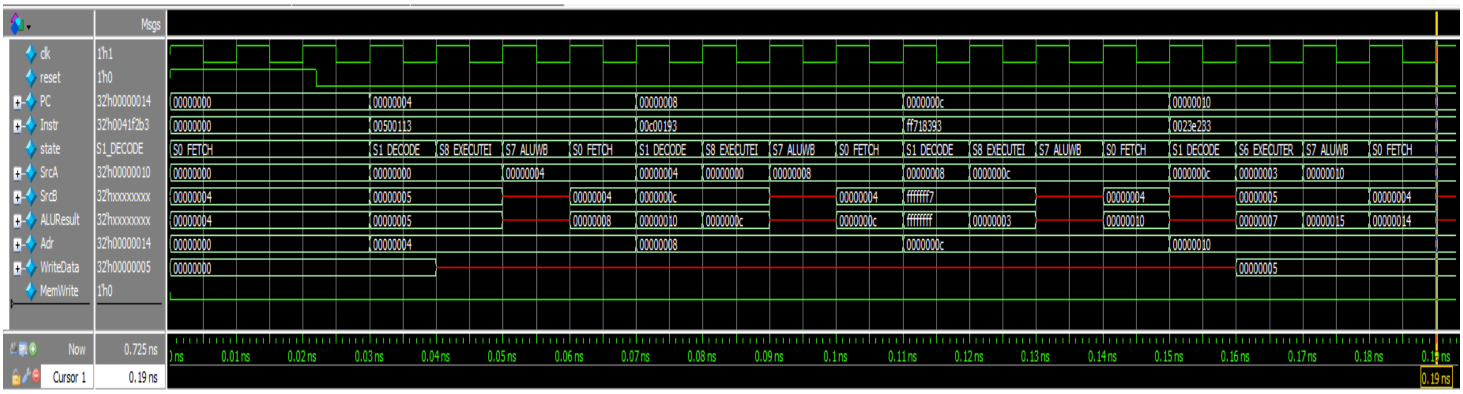
3)

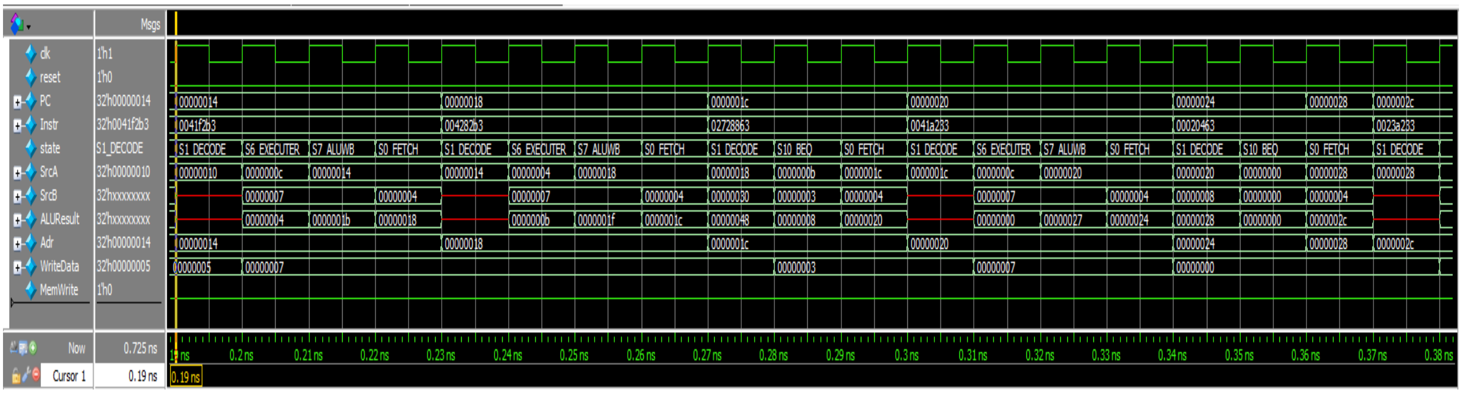


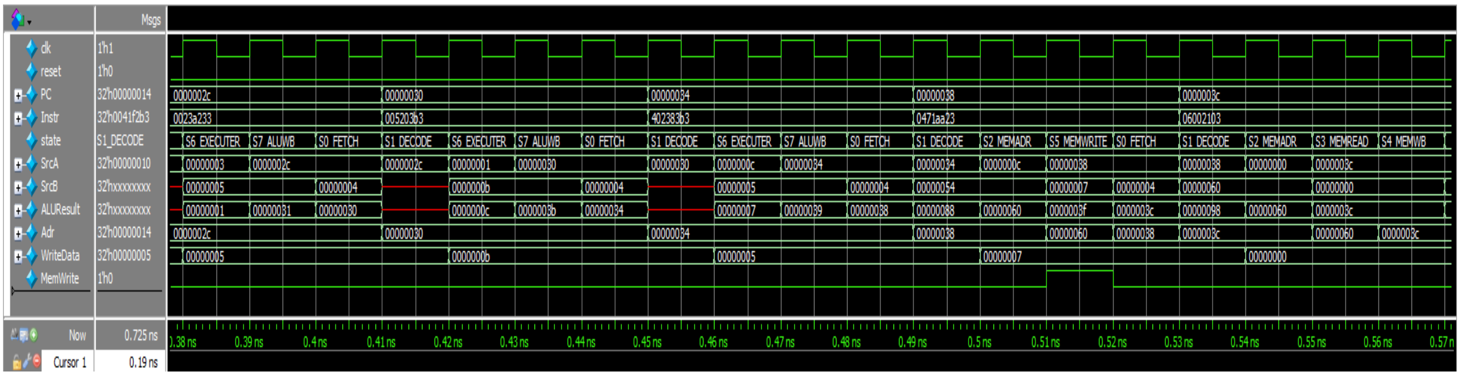
4)

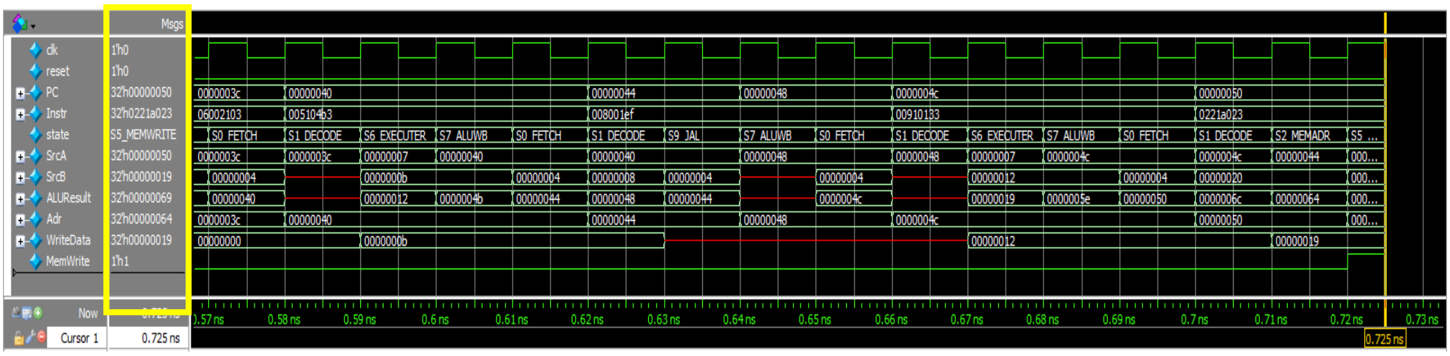
Step	PC	Instr	State	Result	Result Notes
3	00	`00500113`	S0: Fetch	4	PC+4
4	04	""	S1: Decode	X	OldPC+Immediate
5	04	""	S8: ExecuteI	X	ALUResult = x0 (0) + 5 = 5
6	04	""	S7: ALUWB	5	Result = ALUOUT
7	04	""	S0: Fetch	8	PC+4
8	08	`00c00193`	S1: Decode	X	OldPC+Immediate
9	08	""	S8: ExecuteI	X	ALUResult = x0 (0) + 12 = 12 (0xc)
10	08	""	S7: ALUWB	12	Result = ALUOUT
11	08	""	S0: Fetch	C_h	PC+4
12	0c	`ff718393`	S1: Decode	X	OldPC+Immediate
13	0c	""	S8: ExecuteI	X	ALUResult = x3 (12) + (-9) = 3
14	0c	""	S7: ALUWB	3	Result = ALUOUT
15	0c	""	S0: Fetch	10_h	PC+4 *(Starts next instruction)*

5)










```
VSIM 13> run -all
# Simulation succeeded
# ** Note: $stop    : C:/Users/Casper/AppData/Local/quartus_5_mc_processor/riscv_testbench.sv(30)
#    Time: 725 ps  Iteration: 1  Instance: /riscv_testbench
# Break in Module riscv_testbench at C:/Users/Casper/AppData/Local/quartus_5_mc_processor/riscv_testbench.sv line 30
```

https://github.com/Hakan-26/Multicycle-Processor-Full